Measuring Planck's Constant using the Photoelectric Effect

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1 Goal

We produce an experiment that investigates the relationship between photons' wave frequencies and their energies. We configure a system that liberates electrons from the surface of a metal using the photoelectric effect, using lights of several frequencies. By measuring the sensitivity of these electrons' kinetic energies to the frequency of light used to induce their escape from the metal, we determine the factor a photon's frequency must be scaled by to obtain its energy, known as Planck's constant h. As a byproduct of our procedure, we identify the effective work function of the metal the electrons were freed from. We also investigate the relationship between a photocurrent produced and the retarding voltage applied to the photocurrent-carrying electrons.

2 Introduction and Background

The photoelectric effect is a phenomenon during which photons come into contact with the surface of a material and free electrons from the material's potential well [2]. Electrons occupy bound states in the in a metal material with a variety of energy levels—in other words, states at varying depths in the potential well. The maximum energy level is labeled the Fermi energy ϵ [2]. We define the work function of a metal W as the energy required to free an electron at the Fermi energy from its bound state, making it a conduction electron. In other words, the work function of a metal is the positive difference between the level of zero potential energy and the Fermi energy [2].

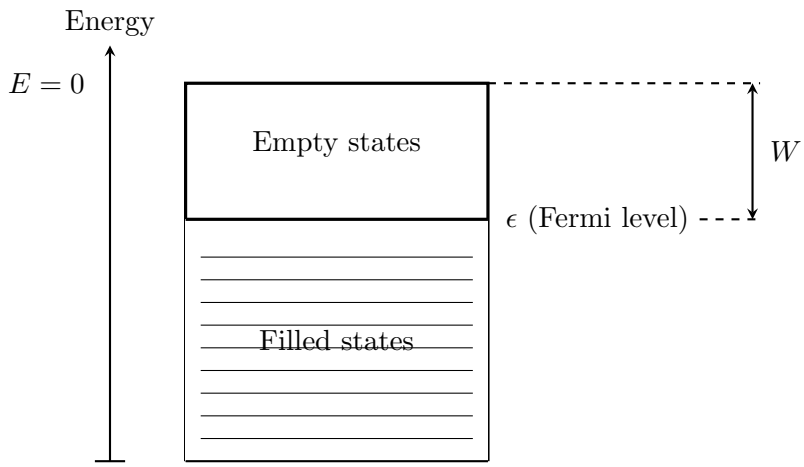

Figure 1: Energy levels in a metal, showing the Fermi level, work function, and filled and empty states.

When a photon frees a bound electron in such a manner, the electron inherits the photon's energy E, equal to the photon's frequency f up to a factor of h [2]:

$$E = hf \tag{1}$$

The frequency of a photon f is related to its wavelength λ and the speed of light in a vacuum c by [1, 2]:

$$c = \lambda f \tag{2}$$

An electric potential difference between an anode and a cathode creates a potential gradient from high potential V_c to low potential V_a, producing a negative voltage jump $V_a - V_c$ from the cathode to the anode [2].

The potential energy of a particle of charge q at a point in space where the electric potential is V_0 is [2]:

$$U_0 = qV_0 \tag{3}$$

The relationship between the potential energy a photoelectron originally at the Fermi level must accrue to be stopped by a potential difference ΔV, the wavelength λ of the photon, the work function W of the metal, the speed of light c, the fundamental charge e, and and Planck's constant h is:

$$-e\Delta V = eV_s = \frac{hc}{\lambda} - W \tag{4}$$

where we introduce a positive quantity V_s, the stopping voltage, such that $V_s = -\Delta V$ [2].

3 Procedure

We establish a cathode-anode pair with the cathode's potential V_c held at the ground potential and the anode's potential V_a at a potential we can adjust using a power supply. Applying (3), electrons

forced to pass from the cathode to the anode against the potential gradient accrue additional positive potential energy $q\Delta V = -e(V_a - V_c)$ upon reaching the anode, where e is the fundamental charge. By conservation of energy principles, the electron's kinetic energy has decreased by $q\Delta V$ during this process. Thus, as we increase V_c, the current measured through the picoammeter decreases as photoelectrons of lower kinetic energies are no longer able to reach the anode; these electrons' kinetic energies are less than or equal to $q\Delta V$.

Before running our experiment to determine Planck's constant, we investigate the relationship between the anode-cathode retarding voltage and the photoelectric current remaining after the photoelectrons cross the voltage jump. We measure the current produced by 570nm light at varying voltages of the anode.

At the point where the current is 0nA, the electrons of the highest kinetic energies are just able to reach the anode. These electrons, having the highest kinetic energies of those in the photocurrent produced, correspond to electrons originally at the Fermi level and requiring the smallest contribution from the photon-supplied energy to escape the well.

We obtain 9 LED lights of different wavelengths. A spectrometer provides an empirical intensity vs. wavelength curve for each light, to which we fit Gaussian models. The Gaussian models' 1σ regions provide the full width at half maximum boundaries of the light's wavelength distribution, while the maximum-intensity wavelength from the points provided by the spectrometer provides the light's wavelength. For the wavelengths of light in our sample set, we configure an initial photocurrent of (5.21 ± 0.02) nA at zero voltage by adjusting the light's intensity. We then increase the voltage gap between the anode and cathode until we achieve the zero-current configuration. We record the stopping voltage V_s corresponding to this zero-current configuration.

This allows us to plot empirical samples of the relationship between $-e\Delta V$ and $\frac{c}{\lambda}$ from (4). The slope of a linear regression model fit to this dataset produces h, our experimental value for Planck's constant. We do not calculate W, attributing an extrapolated intercept appearing during linear regression to W.

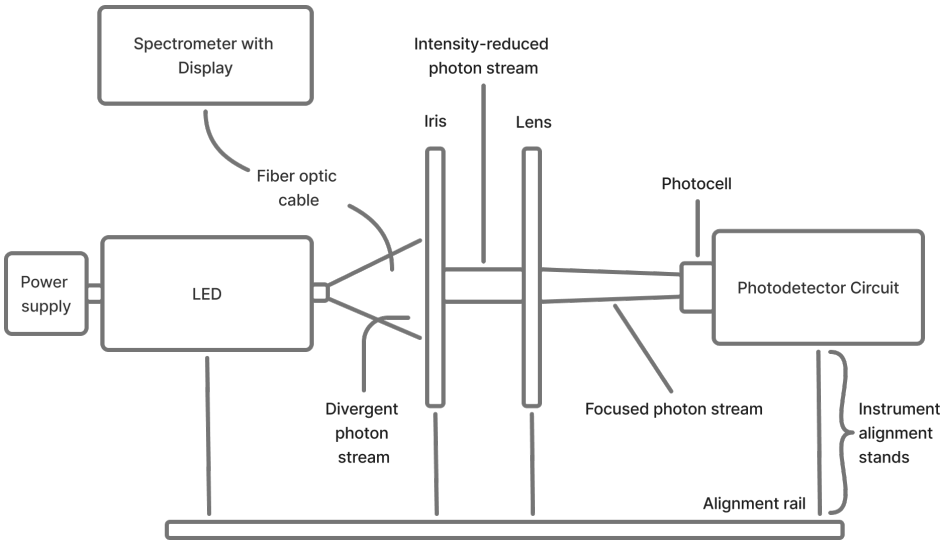

Figure 2: Optical rail system for inducing a photocurrent using LEDs. An LED faces an iris, used to control light intensity. We use a lens to focus the light onto a photocell linked to a photodetector circuit. Instruments are mounted on stands constrained to a rail to allow for alignment adjustments. A spectrometer with a fiber optic cable is used to measure intensity vs. wavelength relationships for light emitted by each LED.

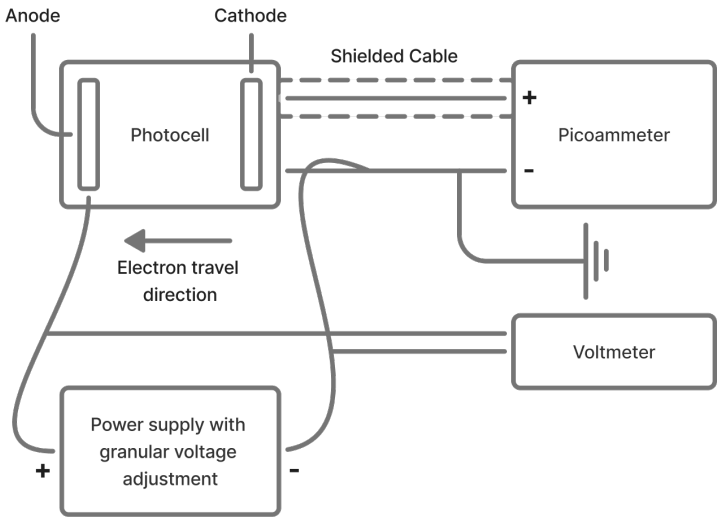

Figure 3: Expanded view of the photodetector circuit. Electrons are liberated from the photocell and flow from the grounded cathode to the voltage-controlled anode. The resulting photocurrent is measured by a picoammeter and a voltmeter provides monitoring of the voltage across the diode [2].

4 Results and Analysis

The voltages and corresponding currents resulting from our preliminary voltage-current experiment are shown in Table 1 and plotted in Figure 5.

Voltage (V), all ± 0.04 V	Current (nA), all ± 0.02 nA
−3.09	68.5
−2.75	66.94
−2.56	65.57
−2.25	64.01
−2.00	62.4
−1.75	60.65
−1.50	58.61
−1.25	55.93
−1.00	51.91
−0.75	46.33
−0.50	38.14
−0.25	28.48
0.00	16.14
0.25	6.23
0.30	3.03
0.40	0.99
0.50	0.26
0.60	0.03
0.70	−0.02
1.00	−0.04
2.00	−0.05
3.00	−0.05

Table 1: The voltages applied to the photoelectrons and the corresponding currents after the photoelectrons cross the voltage jump.

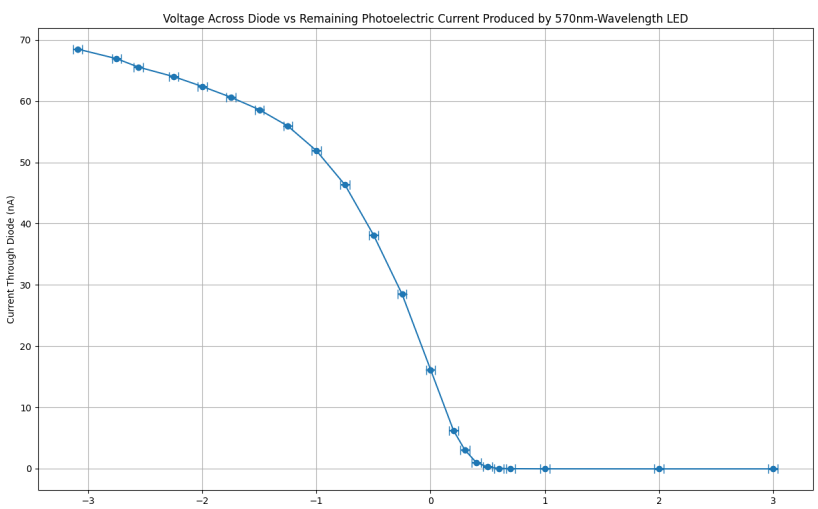

Figure 4: A plot of our measured values of the anode-cathode voltage and the corresponding currents after the photoelectrons cross the voltage jump, from Table 1.

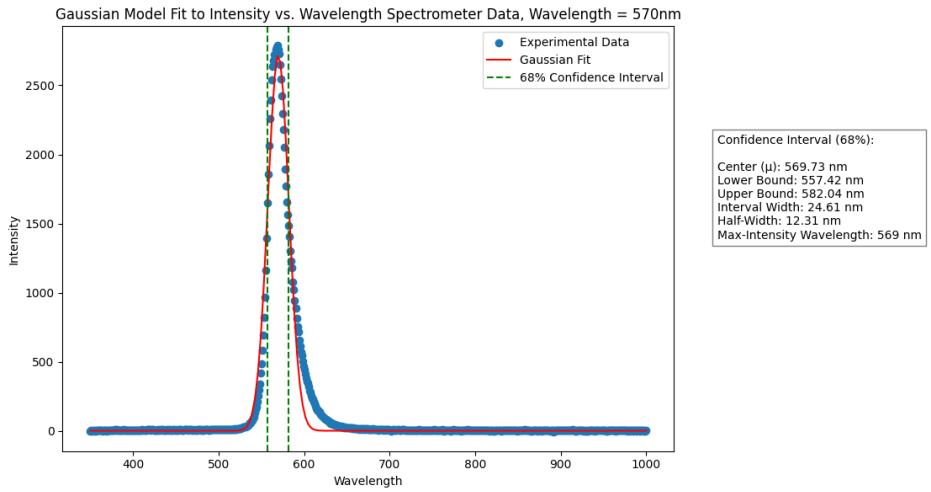

Figure 5: An example of a Gaussian model fit to intensity vs. wavelength spectrogram data. Note that the 68% confidence interval provides double the uncertainty of the maximum-intensity wavelength.

Our measured values of the light wavelengths and the corresponding stopping voltages are shown in Table 2. We plot our measured values of $-e\Delta V$ versus $\frac{c}{\lambda}$ in Figure 6 and fit a linear trend line to the data. We also perform a pointwise analysis of the dataset in table 2, using (4) to compute $h = \frac{\lambda}{c}(W + eV_s)$ for each light using the empirical value for W from the linear regression fit. These results are also shown in Table 2 and plotted in Figure 7.

Max Wavelength (nm)	Stopping Voltage (V_s)	h ($\times 10^{-34}$ J·s)
438.0 ± 3.0	1.2635 ± 0.0001	6.57 ± 0.18
466.0 ± 13.0	1.1400 ± 0.0001	6.68 ± 0.26
506.0 ± 12.0	0.8612 ± 0.0001	6.50 ± 0.25
519.0 ± 14.0	0.8044 ± 0.0001	6.51 ± 0.27
569.0 ± 12.0	0.6306 ± 0.0001	6.61 ± 0.27
592.0 ± 7.0	0.5669 ± 0.0001	6.67 ± 0.25
611.0 ± 7.0	0.4929 ± 0.0001	6.64 ± 0.26
639.0 ± 9.0	0.4073 ± 0.0001	6.66 ± 0.27
650.0 ± 10.0	0.3113 ± 0.0001	6.44 ± 0.28

Table 2: Photoelectric effect measurements: wavelengths of LED lights (with uncertainties from Gaussian fit widths), stopping voltages of Fermi-level photoelectrons, and calculated values of Planck's constant.

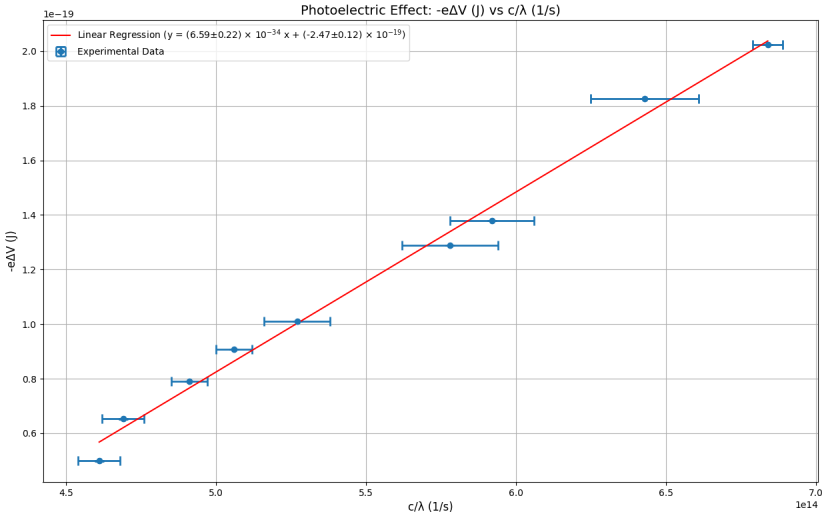

Figure 6: A plot of our measured values of $-e\Delta V$ versus $\frac{c}{\lambda}$, with a linear trend line fitted to the data and the slope of the trend line providing an experimental value for Planck's constant: $h = (6.59 \pm 0.22) \times 10^{-34}$ J·s. The intercept of the trend line provides an experimental value for the work function of the metal: $W = (2.47 \pm 0.12) \times 10^{-19}$ J. Note: this figure does not include the origin.

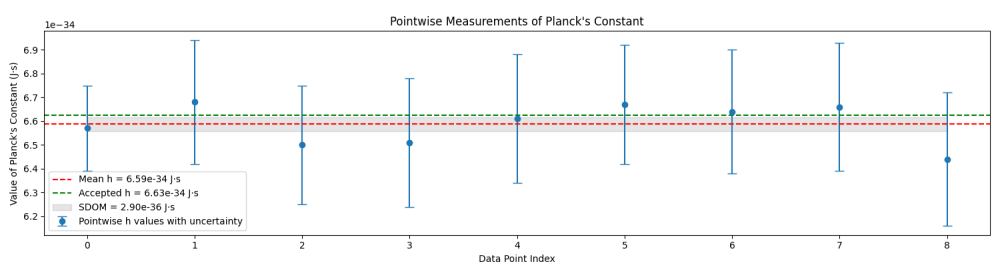

Figure 7: A plot of our measured values of Planck's constant with measurement uncertainties, derived from single data points. The plot indicates the mean value across all measurements and the accepted value of Planck's constant. The mean provides our experimental value.

The value for h obtained from our regression analysis is $h + \sigma_h = (6.59 \pm 0.22) \times 10^{-34}$ J·s, which is consistent with the accepted value of 6.626×10^{-34} J·s to within 1σ. The value obtained from our pointwise analysis is $\bar{h} + \sigma_{\bar{h}} = (6.587 \pm 0.029) \times 10^{-34}$ J·s, which is consistent with the accepted value to approximately 1.35σ. Our value for the work function of the photocell metal W, $(2.47 \pm 0.12) \times 10^{-19}$ J $= (1.54 \pm 0.07)$ eV, is a measure of the effective work function of the metal. This value is "effective" in the sense that the metal surface contains impurities, which alters the states of the electrons on the surface. The work function related to the photoelectric effect is defined under the assumption that the metal surface is free from such impurities [1].
We also note that the work function of our photocell metal does not vary with light intensity, in line with the quantum mechanical model of the photoelectric effect.

We note that the centers of the values we find from linear regression and from SDOM analysis are

consistent with each other, but that the uncertainty of the SDOM analysis is an order of magnitude smaller. We note that SDOM analysis methods often produce uncertainties that might be smaller than what one is comfortable with reporting, and accept the margin of agreement of 1.35σ as an appropriate indicatior that our findings are consistent with the accepted value.
Our disagreement with the accepted value for Planck's constant might be accounted for by several sources of uncertainty: we note that there is a considerable amount of electrical noise in the measurements of the picoammeter, that the initial photocurrent was not stable at its initial value between all wavelengths, and that light sources often had intensity vs. wavelength relationships with significant deviation from a Gaussian distribution. We identify the need to pursue a more controlled method for setting and measuring photocurrents as well as more appropriate models for spectrogram measurements.

5 Conclusion

We obtain the values $h = (6.59 \pm 0.22) \times 10^{-34}$ and $h = (6.587 \pm 0.029) \times 10^{-34}$ for Planck's constant, a fundamental quantity describing the relationship between a photon's frequency and its energy. We successfully apply a procedure involving inducing the photoelectric interaction between lights of several wavelengths and a metal photocell, obtaining values in agreement with the accepted CODATA value. Tangentially, we also determine an effective work function of our photocell metal: (1.54 ± 0.07) eV. Our work also includes an investigation of the relationship between the anode-cathode retarding voltage and the post-gap photoelectric current produced.

6 References

[1] Harris, R. (2008). Modern Physics. Addison-Wesley.

[2] Knight, Randall D, et al. College Physics. Addison-Wesley, 4 Jan. 2016.